

CM0845 Logic Other Topics

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Some Related Subjects

- Logic monism or logic pluralism (Logic or logics?)
- Time matters (algorithmic complexity)
- When you only have a hammer, everything looks like a nail (paradigms of programming)
- Help from the machines (theorem provers)

Logic or Logics?

See slides in the course web page.

Algorithmic Complexity

Comparison of several time complexity functions

$f(n)$	10	50	100
$\log n$	2.3 sec	3.9 sec	4.6 sec
n	10 sec	50 sec	1.7 min
n^2	1.7 min	41.7 min	2.8 h
2^n	17.1 min	358.001 c	4×10^{20} c
3^n	16.4 h	2.3×10^{14} c	1.6×10^{38} c
$n!$	42 d	9.7×10^{54} c	3×10^{148} c

Algorithmic Complexity

Example (3-SAT: An intractable problem)

To determine the satisfiability of a propositional formula in conjunctive normal form where each disjunction of literals is limited to at most three literals.

The problem was proposed in Karp's 21 NP-complete problems.[†]

[†]Karp (1972). Reducibility Among Combinatorial Problems.

Algorithmic Complexity[†]

3-SAT better deterministic algorithmic complexity

Year	Complexity	Reference
2010	$O(1.439^n)$	Kutzkov and Scheder (2010). Using CSP to Improve Deterministic 3-SAT (unpublish).
2008	$O(1.465^n)$	Scheder (2008). Guided Search and a Faster Deterministic Algorithm for 3-SAT.
2004	$O(1.473^n)$	Brueggemann and Kern (2004). An Improved Deterministic Local Search Algorithm for 3-SAT.

[†]Moser and Scheder (2011). A Full Derandomization of Schönning's k-SAT Algorithm.

Algorithmic Complexity

3-SAT simulation

Machines from: www.top500.org

Petaflops (PFs): 10^{15} floating-point operations per second

Date	Machine	PFs	100	150	200
2013-06 2015-10	Tianhe-2	33.86	2.0 sec	16.0 y	4.1×10^7 c
2012-06	Blue Gene/Q	16.32	4.1 sec	33.1 y	8.5×10^7 c
2011-06	K computer	8.16	8.1 sec	66.2 y	1.7×10^8 c

Some Paradigms of Programming

Imperative: Describe computation in terms of state-transforming operations such as assignment. Programming is done with statements.

Logic: Predicate calculus as a programming language. Programming is done with sentences.

Functional: Describe computation in terms of (mathematical) functions. Programming is done with expressions.

Examples

Imperative $\left\{ \begin{array}{l} \text{C} \\ \text{C}++ \\ \text{Java} \end{array} \right.$

Logic $\left\{ \begin{array}{l} \text{CLP(R)} \\ \text{Prolog} \end{array} \right.$

Functional $\left\{ \begin{array}{l} \text{ML} \\ \text{Erlang} \\ \text{Pure} \left\{ \begin{array}{l} \text{Clean} \\ \text{Haskell} \\ \text{Idris} \end{array} \right. \end{array} \right.$

Theorem Provers

See slides in the course web page.

References

- Tobias Brueggemann and Walter Kern (2004). An Improved Deterministic Local Search Algorithm for 3-SAT. Theoretical Computer Science 329.1–3, pp. 303–313. DOI: [10.1016/j.tcs.2004.08.002](https://doi.org/10.1016/j.tcs.2004.08.002) (cit. on p. 6).
- Richard M. Karp (1972). Reducibility Among Combinatorial Problems. In: Complexity of Computer Computations. Ed. by Raymond E. Miller and James W. Thatcher. Plenum Press, pp. 85–103. DOI: [10.1007/978-1-4684-2001-2_9](https://doi.org/10.1007/978-1-4684-2001-2_9) (cit. on p. 5).
- Konstantin Kutzkov and Dominik Scheder (2010). Using CSP to Improve Deterministic 3-SAT. CoRR abs/1007.1166. URL: <https://arxiv.org/abs/1007.1166> (cit. on p. 6).
- Robin A. Moser and Dominik Scheder (2011). A Full Derandomization of Schönning's k-SAT Algorithm. In: Proceedings of the Forty-third Annual ACM Symposium on Theory of Computing (STOC 2011), pp. 245–252. DOI: [10.1145/1993636.1993670](https://doi.org/10.1145/1993636.1993670) (cit. on p. 6).

References

Dominik Scheder (2008). Guided Search and a Faster Deterministic Algorithm for 3-SAT. In: Proc. of the 8th Latin American Symposium on Theoretical Informatic (LATIN 2008). Ed. by Eduardo Sany Laber et al. Vol. 4957. Lecture Notes in Computer Science. Springer, pp. 60–71. DOI: [10.1007/978-3-540-78773-0_6](https://doi.org/10.1007/978-3-540-78773-0_6) (cit. on p. 6).